

PCA

Principal Component Analysis

Noam Yakar

Arizona State University

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What is Principal Component Analysis?

- PCA is a dimensionality reduction technique.
- Used to reduce high-dimensional data into fewer components.
- Preserves variance in the data while simplifying its structure.

Why did I choose PCA?

- PCA is essential in data science and machine learning.
- It helps handle large datasets in an efficient way.
- The technique has practical use in my field of interest, especially for visualizing data.

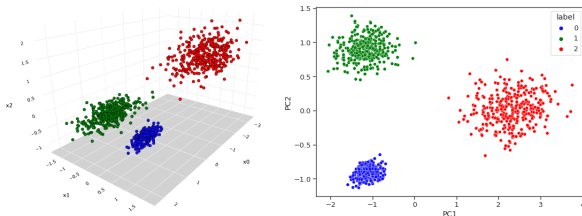
It turns out that most data is messy, and has a lot of features. Take RNA Seq Data, you are measuring the expression of individual cells, you may have **thousands of features**.

So what's the problem with high dimensional data?

- Training complex learners is inherently expensive
- Hard time generalizing
- Can we find a lower dimensional space that maintains certain properties of the data?

Advantages of PCA

- Reduces complexity in large datasets.
- Helps visualize data in 2D or 3D.
- Useful for noise reduction.
- Identifies key components (principal components) driving the variance.



Challenges of PCA

- Not suitable for non-linear data.
- Loss of interpretability of original features.
- Sensitive to scaling of features (requires normalization).
- Can discard important information if not used carefully.

When does PCA fail?

Non linear relationships in data, this is where methods like t-SNE and kernel PCA come in.

Case Study: PCA in Gene Expression Data

- PCA is commonly used in analyzing RNA-Seq data.
- Thousands of gene expressions can be reduced to key features.
- Helps researchers identify patterns in gene expression related to diseases.

- PCA can be combined with other non-linear techniques like t-SNE for better results.
- In big data, it will remain crucial for preprocessing before applying more complex algorithms.
- Potential applications in AI and predictive analytics.

The Spectral Theorem

Let $A \in \mathbb{R}_{n \times n}$ be a real symmetric matrix. The Spectral Decomposition Theorem states that any such A can be decomposed as

$$A = QDQ^T$$

Where Q is an orthogonal matrix, and D is a diagonal matrix. This is a major theorem in linear algebra, that will be used in finding the variance maximizing vectors.

Covariance Matrix I

Let $\mathbb{X} \in \mathbb{R}^{n \times d}$ be our data matrix. It has n observations, d features.¹ Each of the d features can be represented as a random variable X_i , and each X_i has a finite mean and variance.

Define

$$S = \begin{bmatrix} \text{var}(X_1) & \text{cov}(X_1, X_2) & \cdots & \text{cov}(X_1, X_d) \\ \text{cov}(X_2, X_1) & \text{var}(X_2) & \cdots & \text{cov}(X_2, X_d) \\ \vdots & \vdots & \ddots & \vdots \\ \text{cov}(X_d, X_1) & \text{cov}(X_d, X_2) & \cdots & \text{var}(X_d) \end{bmatrix}$$

This covariance matrix is symmetric, and positive semi-definite, which means it is diagonalizable!

¹ $\mathbb{X}$ has n -independent copies of the d dimensional feature vector

Covariance Matrix II

Recall our goal is to find special vectors that maximize the variance of our data. It turns out that the variance of our data along a vector a can be written as

$$a^T S a$$

Now what's so special is that the vectors that maximize the variance in our data turn out to be the eigenvectors of our covariance matrix.²

These eigenvectors can be found in the matrix P !³

²Ask me about the proof if you're interested, it involves the spectral theorem

³these are called the principal components

1. Start with your data $\mathbb{X} \in \mathbb{R}^{n \times d}$
2. Compute the empirical covariance matrix S
3. Choose $k < d$, and set P to the first k eigenvectors of S ⁴
4. Project the original data onto the first k eigenvectors

⁴How to choose this k ?

- PCA is a powerful technique for simplifying high-dimensional data.
- It helps reveal underlying structures and patterns.
- Despite its limitations, PCA continues to be an essential tool in data science.

- Jolliffe, I. T. (2002). Principal Component Analysis.
- Setosa.io. "Eigenvectors and Eigenvalues Visualization."
setosa.io/ev/eigenvectors-and-eigenvalues
- Hastie, T., Tibshirani, R., Friedman, J. (2009). The Elements of Statistical Learning.